

of *Igf-2* transcription^{1,8}. Intriguingly, unlike the loss of *Igf-2r* activity, *Igf-2* activity is lost with the weaker *MTaseⁿ* mutant allele, suggesting that there is a greater sensitivity of the *Igf-2* upstream region and the *H19* gene to methylation changes. Further investigation should resolve whether or not there is a direct relationship between the imprinting of the *H19* and *Igf-2* genes.

Demethylation

Another question — that of whether differential methylation patterns are inherited from the germ line — is pertinent because such imprints must survive the striking genome-wide demethylation that accompanies pre-implantation development^{11–13}. Overall DNA methylation is exceedingly low in blastocysts, which may be crucial for establishing pluripotential epiblast cells. It might be informative to test whether low-level transcription of imprinted genes occurs in blastocysts and, if it does, to see whether both parental alleles of imprinted genes are expressed. Later, with the onset of gastrulation, *de novo* methylation occurs around day 6.5 of development and includes methylation of the paternal *H19* promoter^{7,14}. Whether the primordial germ cells escape that methylation is a moot point, but they have the capacity to erase methylation and establish the new germline pattern^{15,16}.

The *Igf-2r* intronic region is perhaps the most clear-cut example of the inheritance of methylated sites from oocytes that remain intact in blastocysts^{3,14}. Some sites in the upstream region of the *Igf-2* gene are also methylated, but in both sperm and oocytes; although many of these sites undergo demethylation in pre-implantation embryos, one of the paternally inherited sites remains methylated at least in some of the cells¹⁴. In the *H19* gene, no sites have yet been identified that are inherited as differentially methylated from gametes. At present it is not possible

to say unequivocally if DNA methylation is the signal that specifies the germline imprint for any or all imprinted genes. But the inability of methyltransferase mutant embryos to maintain normal regulation of imprinted genes confirms the view that DNA methylation is a key event in the imprinting process.

Of the three imprinted genes, the function of *H19* gene, which appears not to encode a protein, remains enigmatic, although recent evidence indicates that the *H19* RNA may have tumour-suppressor activity¹⁷. We already know that imbalanced expression of imprinted genes is lethal. Embryos with the maternal disomy for chromosome 7 have excess *H19* and no *Igf-2* (just like the methyltransferase mutant embryos), and *Tme*

mutation embryos lack the functional *Igf-2r* gene; both are lethal^{2,7}. This means that the methyltransferase mutants cannot survive because of the perturbation of the three genes. But there may be more widespread disturbance of gene regulation, and these mutant mice could prove useful for investigating the extent to which DNA methylation controls early development. More notably, though, it should now also be feasible to investigate the influence of DNA methylation in the germ line and the consequences for genomic imprinting. □

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MATERIALS SCIENCE

Confusion by design

A. Lindsay Greer

GLASSY metals mostly teeter on the verge of viability. They are difficult to produce in the first place because they require such high cooling rates, and they show an ungrateful tendency to undo the good work by crystallizing on heating, hindering the customary hot-forming processes and limiting applications. This has been frustrating, to put it mildly, for those wishing to exploit the unusual properties of the materials, so last month's news of the work of Peker and Johnson¹ should come as music to their ears. By including so wide a range of metallic elements that the alloy was essentially too confused to crystallize, they have devised an exceptionally stable, user-friendly glass.

Since the first reports of metallic glasses produced from the liquid², their formation has been associated with ultra-high cooling rates, typically over 10⁵ kelvin per second, and this has restricted production to forms with very thin cross-section (ribbon, wire or droplets). This isn't an inescapable requirement: there have been a few notable exceptions, for instance Pd₄₀Ni₄₀P₂₀ glass (all compositions are in atomic per cent), for which bulk sections 10 millimetres across could be produced at a cooling rate of only 1 kelvin per second — the catch being that the alloy must be kept very clean in order to inhibit heterogeneous nucleation³.

The field really took off, though, with the demonstration that excellent glass-forming ability could be found even without noble metals (which are costly) or metalloids (whose inclusion somewhat detracts from the idea of having a fully metallic glass, and may lead to brittleness). The group of Inoue and Masumoto⁴ showed that alloys of the family Ln–Al–TM (where Ln represents a lanthanoid

and TM a transition metal; an example is La₅₅Al₂₅Ni₂₀) could be cast to form cylinders and sheets a few millimetres in cross-section. The group went on to show that it was not essential for a lanthanide to be the base element (for example, they obtained good results from Mg–TM–Ln alloys such as Mg₆₅Cu₂₅Y₁₀)⁵, or indeed to be present at all (as in Zr–Al–TM alloys)⁶.

What was going on? In the latter alloy type⁶, the Japanese group showed that the best results could be obtained with more than three elements, stirring in, for example, two different transition metals (Zr₆₅Al_{7.5}Ni₁₀Cu_{17.5}). Considering variations in atomic radius, they suggested that a mixture of elements leads to dense packing in the liquid state, and the resulting stability favours glass formation over crystallization.

Peker and Johnson have now taken the logical next step, with the addition of beryllium. Their composition — Zr_{41.2}Ti_{13.8}Cu_{12.5}Ni_{10.0}Be_{22.5} — sets the pattern for anomalously easy glass formation in alloys, and its complexity is just the key to its success.

It seems to have been widely accepted by those working on metallic glass formation that a 'confusion principle' should apply: the more elements involved, the lower the chance that the alloy can select viable crystal structures, and the greater the chance of glass formation. So why have there been so few attempts to exploit the idea? The answer, it seems, is that 'two's company, three's a crowd' to those wishing to analyse fully the thermodynamics and kinetics of alloy transformations — the more elements, the more difficult the analysis.

The recent results show that the principle, when applied with care, works very

- Li, E., Beard, C. & Jaenisch, R. *Nature* **366**, 362–365 (1993).
- Barlow, D. P., Stöger, R., Herrmann, B. G., Saito, K. & Schweifer, N. *Nature* **349**, 84–87 (1991).
- Stöger, R. *et al.* *Cell* **73**, 61–71 (1993).
- De Chiara, T. M., Robertson, E. J. & Efstratiadis, A. *Cell* **64**, 849–859 (1991).
- Bartolomei, M. S., Zemel, S. & Tilghman, S. M. *Nature* **351**, 153–155 (1991).
- Sasaki, H. *et al.* *Genes Dev.* **5**, 1843–1856 (1992).
- Ferguson-Smith, A. C., Sasaki, H., Cattaneach, B. M. & Surani, M. A. *Nature* **362**, 751–754 (1993).
- Bartolomei, M. S., Webber, A. L., Brunkow, M. E. & Tilghman, S. M. *Genes Dev.* **7**, 1663–1673 (1993).
- Bird, A. P. *Nature* **321**, 209–213 (1986).
- Li, E., Bestor, T. H. & Jaenisch, R. *Cell* **69**, 915–926 (1992).
- Monk, M., Boubelik, M. & Lehnert, S. *Development* **99**, 371–382 (1987).
- Howlett, S. K. & Reik, W. *Development* **113**, 119–127 (1991).
- Kafri, T. *et al.* *Genes Dev.* **6**, 705–714 (1992).
- Brandeis, M. *et al.* *EMBO J.* **12**, 3669–3677 (1993).
- Chaillet, J. R., Vogt, T. F., Beier, D. R. & Leder, P. *Cell* **66**, 77–83 (1991).
- Ueda, T. *et al.* *Development* **116**, 831–839 (1992).
- Hao, Y., Crenshaw, T., Moulton, T., Newcomb, E. & Tycko, B. *Nature* **365**, 764–767 (1993).

effectively. The most 'confusing' elements will be those that differ most from each other in size, and here lies the key to the use of beryllium: it is the smallest metallic atom. Its addition should offer extra scope for the design of other glass-forming compositions, allowing elements to be selected for properties such as magnetism and not just for their glass-forming ability.

The magnitude of the advance for metallic glasses can be gauged most directly from sample size: a conventional alloy made glassy by ultra-rapid quenching is limited to a section thickness of less than 0.1 millimetres, whereas Peker and Johnson, using simple water quenching (cooling at 10 kelvin per second or less), have made fully glassy rods 14 millimetres in diameter, and this is not regarded as an upper limit. With samples of such a size, whole new areas of application become available to amorphous alloys, allowing us to exploit desirable properties such as high strength, corrosion resistance and soft magnetism.

Peker and Johnson have already shown that it is possible to cast and machine small components. The unusual resistance to crystallization would make possible the shaping of sheet metal by viscous flow at elevated temperature. This perfect superplasticity is widely exploited in the forming of conventional oxide glasses, but would be a considerable novelty for metallic alloys.

Besides being of interest for large samples, the formability of the new glassy alloys may be useful for very small components. Microforming of gear wheels with feature resolution of 1 micrometre has been reported⁷, the advantage of the glass being its homogeneous isotropic deformation. And a bonus is that the new materials can be consolidated without crystallizing; making sizeable components from the flimsy ribbons or droplets of the old method was an art in itself.

With careful design of compositions, easy glass formation should be possible in quite a variety of multicomponent alloy systems. Given the sheer complexity of analysis and calculation of phase diagrams and kinetics for multicomponent systems such as these, it seems likely that metallic glasses have a few surprises still in store. □

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Hot lasers, cool molecules

Herschel Rabitz

HOT molecules may be generated easily as the result of chemical reactions or through external heating. Cooling such molecules (reducing their vibrational temperature) allows a detailed look at molecular dynamics and might ultimately help chemists to realize their dream of controlling reaction pathways at this smallest of scales. Bartana and co-workers have come up with a scheme for using finely controlled and designed laser pulses to do just that (*J. chem. Phys.* **99**, 196–210; 1993).

The authors consider an ensemble of molecules vibrationally excited in their ground electronic state. A traditional approach to molecular cooling might be to remove the population amongst the most vibrationally excited levels and transfer that population elsewhere, perhaps to an excited electronic state. In contrast, Bartana and colleagues consider a novel cooling procedure which does not change the overall population in the ground state, but rather rearranges it through coherent radiative coupling with an excited electronic state.

The best analogy is with an everyday refrigerator, which employs a working fluid to carry heat away from the cold region (the refrigerator interior) and dump it into a waste heat sink (the surrounding kitchen). The molecular-scale quantum mechanical refrigerator operates under the same principles, but now the desired cold 'region' is the ground electronic state of the molecule, the working fluid is a coherently designed optical field, and the heat sink is an excited electronic state.

Population

A carefully designed optical field with coherent phase structure can, say the authors, maintain the overall population in the ground electronic state, while rearranging it effectively to lower the temperature in that state. The molecule's temperature is characterized by the entropy or population dispersion in the ground electronic state; a lowering of the temperature corresponds to a lowering of the entropy by reducing the dispersion amongst the vibrational states. Cooling would ultimately stop when the population had been driven into a single vibrational level.

This ground-state cooling is achieved at the expense of heating a small fraction of the population in an electronic excited state. The coherent radiation of the field provides energy to lower the entropy on the ground-state surface, while raising the corresponding entropy and temperature on the excited-state surface. Except for an initial seed population introduced in the excited state, the suggested cooling

mechanism involves no net transfer out of the ground electronic state.

No claim is made that practical molecular refrigeration will be achieved in this fashion. But this research is a clever example of the latest theoretical schemes for using tailor-made optical pulses to control events at the most intimate atomic and molecular scales.

Control

The dream of molecular control goes back to the early 1960s, when the idea was born of using lasers to manipulate molecular dynamics and/or associated electronic processes, with the goal of selectively breaking chemical bonds (see R. N. Zare, *Nature* **365**, 105; 1993). A long era of interesting but somewhat misguided research followed. Theory largely focused on various issues of molecular dynamics, but never quite came to grips with the problem of designing the control laser. Meanwhile, laboratory researchers ploughed ahead using all manner of existing lasers without proper guidance from theory, and with an inevitable lack of success. Now, with advances on both the theoretical and experimental fronts (for a review, see W. S. Warren, H. Rabitz and M. Dahleh, *Science* **259**, 1581–1589; 1993), the general consensus is that the research is about to bear fruit.

The key theoretical advance made in the past few years is understanding that control of molecular dynamics is inherently a matter of manipulating quantum interference processes. The generic problem may be posed as starting with the molecule in quantum state ψ_i , and steering it to final state ψ_f , in such a way that some molecular objective is met, be it bond-breaking, localized energy deposition or molecular cooling. For experimentalists, the introduction of optimal control techniques at last gives them the tools to address the design of the optical fields. Optimal control theory has its roots in operations research, going back several decades, and the topic has proved to be enormously successful for treating everyday engineering design problems (for example in the flight of aeroplanes). Theorists are now showing that these concepts can just as well be applied to guiding and manipulating the motion of molecules. The past few years have seen frenetic exploration of the concepts of quantum control theory, along with a variety of preliminary illustrations of the control of rotational, vibrational, reactive and electronic degrees of freedom (in some cases simultaneously).

Early approaches largely used continuous-wave or possibly pulsed lasers with

1. Peker, A. & Johnson, W. I. *Appl. Phys. Lett.* **63**, 2342–2344 (1993).
2. Klement, W. *et al.* *Nature* **187**, 869–870 (1960).
3. Drehman, A. J. & Greer, A. L. *Acta Metall.* **32**, 323–332 (1984).
4. Inoue, A. *et al.* *Mater. Trans. JIM* **31**, 425–428 (1990).
5. Inoue, A. *et al.* *Mater. Trans. JIM* **33**, 937–945 (1992).
6. Zhang, T. *et al.* *Mater. Trans. JIM* **32**, 1005–1010 (1991).
7. Saotome, Y. *et al.* *Proc. 8th Int. Conf. on Rapidly Quenched Metals, Mater. Sci. Engng A* (in the press).